

Microgravity accelerates skeletal muscle degeneration: Functional and transcriptomic insights from an ISS muscle lab-on-chip model

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SUMMARY

Microgravity accelerates skeletal muscle degeneration, mimicking aspects of aging, yet its effects on muscle cell function remain underexplored. Using a muscle lab-on-chip model onboard the International Space Station (ISS), we examined 3D-bioengineered myobundles derived from young and older adult donors under microgravity. Electrical stimulation applied intermittently to the myobundles revealed reduced contraction magnitude in microgravity and decreased protein levels of myosin heavy chain 7, a main isoform in slow-twitch muscle fibers. Transcriptomic profiling revealed active myogenesis across ground and spaceflight samples, but younger electrically stimulated myobundles displayed enhanced mitochondrial-related gene expression in microgravity, while older and non-electrically stimulated myobundles were less responsive. Comparative analysis between young and older derived myobundles identified 86 muscle-specific age-associated genes altered in microgravity, linked to inflammation, mitochondrial dysfunction, and cellular stress. These findings highlight a unique age-related molecular response in microgravity and underscores electrical stimulation as a potential countermeasure. These insights advance our understanding of muscle aging and degeneration in microgravity, guiding future therapeutic strategies.

INTRODUCTION

With the rise of human space exploration, research has focused on understanding how space travel affects human health, the progression of diseases, and the body's mechanisms for maintaining homeostasis. Extended spaceflight missions cause various health issues, with multisystemic dysfunction being a primary concern (Krittawong et al., 2022; Vernikos and Schneider, 2010). The musculoskeletal system is particularly vulnerable to the effects of spaceflight. Prolonged muscle disuse and the absence of gravity-induced mechanical loading result in rapid muscle atrophy (Comfort et al., 2021; Juhl et al., 2021; Lee et al., 2022). It is well established that skeletal muscle adapts to changes in its mechanical environment by altering gene expression (Leuchtmann et al., 2021; Powers and Schrager, 2022; Tidball, 2005). Accelerated muscle atrophy occurring during spaceflight has been proposed to be a relevant model for understanding sarcopenia, the progressive muscle loss primarily driven by aging (Cannavo et al., 2022; Larsson et al., 2019). Astronauts exposed to microgravity experience up to a 30% reduction in skeletal muscle mass and strength within just 1 month of spaceflight (Williams et al., 2009). This decline not only hinders astronauts' ability to perform essential tasks aboard the International Space Station (ISS) but also increases their risk of injury upon returning to Earth's gravity. Without daily, intensive exercise, muscle mass in astronauts spending 6 months in microgravity would re-

semble that of an 80-year-old (Trappe et al., 2009). Investigating the mechanisms behind spaceflight-induced muscle degeneration offers valuable insights into processes underlying age-related muscle degeneration on Earth.

Microgravity affects biology not only at the tissue level *in vivo*, leading to muscle atrophy, but also in a cell-autonomous manner. The growing interest in studying the effects of microgravity on cells has spurred the development of innovative lab-on-chip technologies to facilitate these investigations. Previously, we demonstrated the utility of our 3D skeletal muscle lab-on-chip model for studying muscle biology and stress responses in an autonomous Space Tango CubeLab designed for spaceflight (Parafati et al., 2023). In the experiment, donor-derived myoblasts undergo myogenesis, fusing and responding to chemical cues to form functional, mature human myobundles. Studies have shown that electrical stimulation, which simulates muscle contraction during exercise, enhances the alignment and responsiveness of muscle fibers (Khodabukus et al., 2019; Nikolić et al., 2012; Zhao et al., 2020). Here, we provide an analysis from a subsequent set of tissue chips sent to the ISS in a modified CubeLab in which we cultured the myotubes in the absence or presence of applied electrical stimulation and observed contractile transcriptomic responses as measured by the enriched signaling pathways. These results advanced our muscle tissue chip platform.





The current study investigated how microgravity influences myocyte contractile function and gene expression, and the protective effects of electrical stimulation in 3D-engineered myobundles derived from muscle biopsies of younger and older participants. We choose these participants as a study paradigm to identify new age-related transcriptional signatures associated with muscle cell exposure to microgravity. The myobundles were derived from the same cell stocks for consistency with our previous mission, and based on our earlier results (Parafati et al., 2023), we hypothesized that the younger donor-derived myobundles would exhibit pronounced transcriptional adaptations to microgravity compared to myobundles derived from older donors and that electrical stimulation would partially mitigate these effects.

While our model does not incorporate the traditional mechanical stretch stimulus, our observation of alterations in gene expression with electrical stimulation is consistent with myotube differentiation. Furthermore, enrichment analysis revealed key signaling networks associated with microgravity exposure in myobundles. These findings advance our understanding of muscle health in both space and aging contexts and pave the way for developing targeted countermeasures against muscle atrophy caused by spaceflight and sarcopenia on Earth.

RESULTS

Skeletal muscle tissue chip payload integration for spaceflight

Skeletal muscle myobundles were derived from precursor cells isolated from muscle biopsies of young active (YA) and old sedentary (OS) participants, as previously described (Giza et al., 2022). The cells were embedded in a Matrigel-collagen hydrogel mixture and cultured in microfluidic chips compatible with the spaceflight hardware as previously described (Parafati et al., 2023). An advancement in this mission compared to our previous study was the incorporation of electrical stimulation (E-Stim) for a separate subset of YA and OS-derived myobundles to simulate muscle contraction over 1 week on orbit. The remaining myobundles were not electrically stimulated (No E-Stim), enabling a comparative analysis of global gene expression changes across spaceflight and ground control conditions as well as between E-Stim and No E-Stim treatments.

To facilitate electrical stimulation, we modified the tissue chip design by fabricating grooves in the polydimethylsiloxane (PDMS) structure to secure platinum electrodes along the outer edge of the media channel, positioned parallel to the aligned myobundles (Figure 1A). The chips were integrated into a custom fluid han-

dling manifold that housed eight tissue chips with electrodes (four YA and four OS) on one side, connected to a pulse board, and eight chips without electrodes (four YA and four OS) on the opposite side (Figure 1B). A thermal plate with an open region aligning with the myobundles maintained the culture temperature at 37°C (Figure 1C). The system also included a camera-microscope unit in the CubeLab lid for real-time imaging of myobundle contractions and an LED array underneath the manifold for illumination. This setup enabled autonomous, in-orbit assessment of myobundle contractile function, with parallel ground controls.

The myobundles underwent maturation using an 8-day growth and differentiation protocol prior to launch, consistent with our previous mission (Parafati et al., 2023). Pre-flight, the tissue chips were validated for electrical stimulation responsiveness and reproducibility of peak contractile force. Myobundles underwent twice-daily 30-min electrical pulses at 2 Hz for 7 days, while non-electrode chips served as No E-Stim controls. Contractile activity was analyzed using digital image correlation (DIC) to quantify pixel displacement within a defined region of interest (Figures S1A). The average displacement magnitude for YA- and OS-derived myobundles was not significantly different (Figure S1B). To further characterize the 3D myobundles prior to flight, we assessed viability on day 21, following 7 days of E-Stim, using a live cell fluorescence dye and confirmed cell viability after 12 days of culture in the CubeLab (Figure S1C). Importantly, both YA and OS cohorts showed a similar level of cell composition. Furthermore, α -actinin staining revealed well-organized sarcomere structures, including Z-lines, typical of mature, cross-striated muscle tissue especially after E-Stim (Figure S1D).

Following protocol validation, the myobundles were prepared for spaceflight according to the experimental timeline outlined in Figure 1D. The payload launched from Kennedy Space Center and was installed on the ISS within 3 days. Automated protocols were initiated, including fluid exchanges every 6 h for all 16 tissue chips and electrical stimulation every 12 h for the eight electrode-equipped chips. After the final electrical pulse sequence, RNALater was perfused to each chip (4 chips per treatment group) within 4 h to preserve RNA integrity, and the samples were frozen for return to Earth. Three chips per treatment were used for RNA sequencing (RNA-seq) analysis, and the fourth chip was used for protein extraction to measure myosin heavy chain 7 (MYH7) levels. A parallel ground study was conducted asynchronously postflight in the same CubeLab hardware within a biosafety hood at the University of Florida. Experimental conditions and protocols were synchronized with the spaceflight telemetry to ensure comparability.

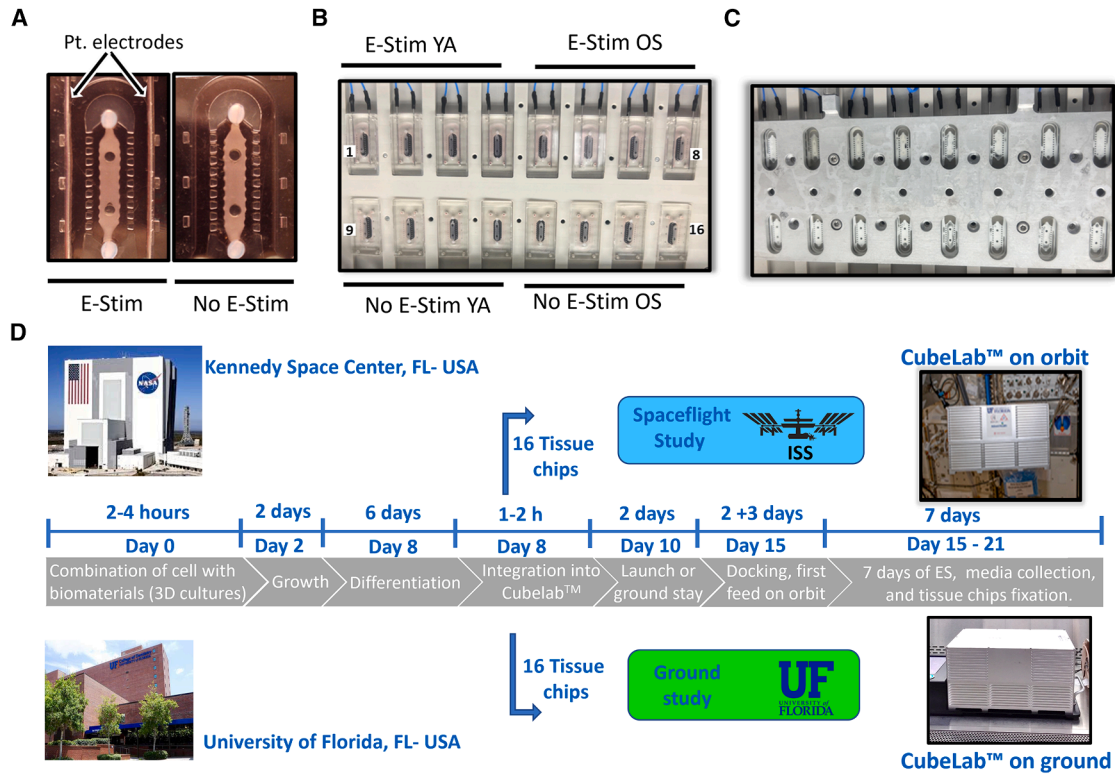


Figure 1. Skeletal muscle payload integration

(A) Skeletal muscle tissues in a custom microfluidic chip with (left) or without (right) platinum electrode leads.
(B) Manifold layout of 16 tissue chips. The top row from left to right includes stimulated young active (E-Stim YA) and old sedentary samples (E-Stim OS). The bottom row from left to right includes non-electrode stimulated young athletic (No E-Stim YA) and old sedentary samples (No E-Stim OS).
(C) Tissue chips are viewable through the cutout for imaging.
(D) Payload time course for the experiment performed on the ISS and asynchronously on ground.

Impact of donor age, E-Stim, and microgravity on myogenesis

Hydrogel-laden myoblasts proliferate, differentiate, and fuse to form mature myobundles within 24 h. To compare the effects of E-Stim versus No E-Stim on myobundle formation, RNA was isolated postflight and found to be in high quantity and quality (Table S1). RNA-seq was performed on myobundles harvested on day 21 (YA and OS, ground, and flight) and compared to a reference group of myobundles formed on day 2 (YA and OS, ground) to identify differentially expressed genes (DEGs) associated with myogenesis. The analysis revealed gene expression profiles indicative of robust muscle differentiation in both age groups (Figures S2 and S3).

Here, we describe results on how E-stim, age, and microgravity impact gene signatures related to differentiation, fusion, and myobundle formation. In No E-Stim YA-derived myobundles, spaceflight significantly increased the number of DEGs compared to ground controls, as visualized by volcano plots (Figures S2A and S2B). Conversely, No E-Stim

OS-derived myobundles exhibited similar gene expression profiles on ground and in space (Figures S2C and S2D). E-Stim YA-derived myobundles also showed a greater number of DEGs in spaceflight compared to ground (Figures S3A and S3B), while E-Stim OS-derived myobundles followed the trend of similar response on ground and in space (Figures S3C and S3D). These findings suggest that both spaceflight and E-Stim induced a more pronounced transcriptional response in YA-derived myobundles compared to OS-derived myobundles. Gene Ontology (GO) enrichment analysis across both age groups identified common pathways associated with muscle cell development and myogenesis, with shared biological processes (BPs), cellular components (CCs), and molecular functions (MFs) across treatments (Figures S2E–S2H and S3E–S3H). GO terms such as muscle contraction, sarcomere organization, Z-disk, T-tubule, structural muscle constituents, and actin filament binding were enriched, confirming substantial transcriptional differentiation between myoblasts and myobundles regardless of E-stim, age, or microgravity exposure.



DEGs: Day 21 myobundles vs. proliferating myoblasts at Day 2

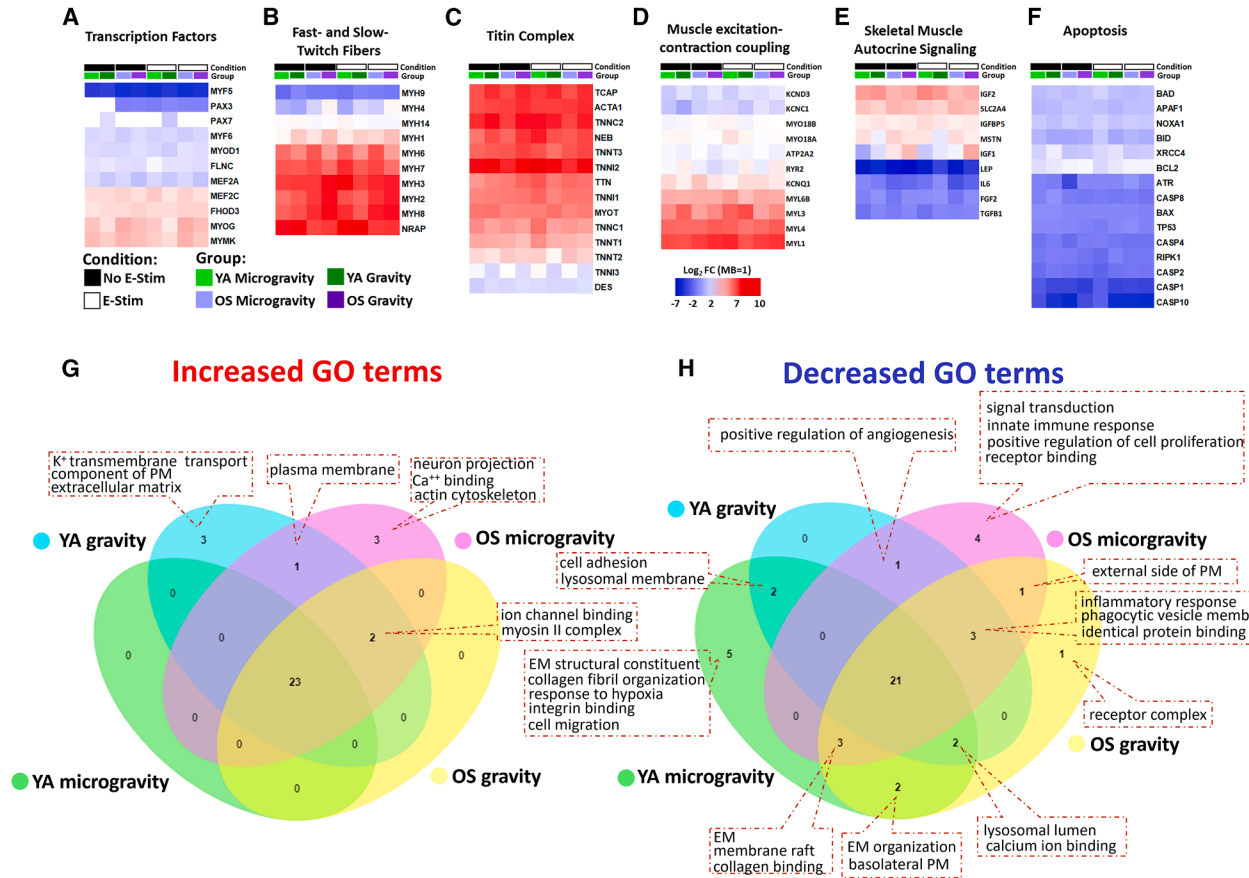


Figure 2. Muscle differentiation comparing day 21 myobundles vs. proliferating myoblasts at day 2

(A–F) Comparisons of DEG changes for No E-Stim (black bars) and E-Stim (white bars) in microgravity (YA light green bars and OS light purple bars) and gravity (YA dark green bars and OS dark purple bars) as heatmap log₂fold change (FC) mean values.

(G and H) Venn diagrams show common and unique GO terms identified in flight (green and purple circles), on ground (light blue and yellow circles), and common to both.

Canonical skeletal muscle transcripts exhibited distinct patterns denoting maturation across conditions (Francetic and Li, 2011; Tripathi et al., 2011; Valdez et al., 2000). Genes promoting myofiber maturation and maintenance (e.g., *MEF2C*, *MYMK*, and *MYOG*) had increased expression in day 21 myobundles compared to day 2 myoblasts, while early differentiation markers (e.g., *MYOD1*, *MYF6*, and *MYF5*) had decreased expression (Figure 2A). Quiescence markers *PAX3* and *PAX7* showed minimal expression across conditions. Muscle-specific MYH genes associated with type I slow-twitch fibers (e.g., *MYH3*, *MYH6*, *MYH7*, and *MYH8*) had consistently increased expression in myobundles versus myoblasts (Figure 2B). Nebulin-related anchoring protein, a key component for myofibril assembly, was significantly increased in all groups (Lu et al., 2008). Highest expression was observed in contractile genes forming the titin complex (e.g., *TNNC3*, *ACTA1*,

TNNT3, and *TNNI2*) and *TCAP*, a mechanical stretch sensor (Figure 2C) (Krüger and Kötter, 2016). Regulatory myosin light chains involved in Ca²⁺-dependent contractile regulation were also expressed (Figure 2D), supporting the predominant formation of type I fibers. Autocrine signaling genes (e.g., *LEP*, *IL6*, *FGF2*, and *TGFB1*) had decreased activity across conditions (Figure 2E). However, *IGF-1*, promoting muscle mass, decreased, and *MSTN*, a negative regulator of muscle growth, increased in YA-derived myobundles exposed to spaceflight, suggesting an age-dependent response to microgravity. Importantly, apoptosis-related genes and programmed cell death genes were decreased supporting myobundle cell health (Figure 2F). In summary, these results indicate that the microgravity environment including launch conditions supports myofiber maturation and maintenance and suppresses cell death during myogenesis.



Further supporting that both YA- and OS-derived myobundles undergo comparable myogenesis under spaceflight and ground conditions, GO enrichment analysis revealed that nearly all gene activity pathways were shared across treatment groups (Figures 2G and 2H). Pathways associated with muscle development and function were increased, while those related to mechanosensing components and secretory processes were decreased. These findings indicate that both YA and OS-derived myobundles exhibited a robust capacity for growth and differentiation in microgravity. Importantly, spaceflight conditions did not impair myobundle differentiation or maturation, underscoring the reliability for using our model for endpoint analyses across the groups and different gravitational environments.

E-Stim-conditioned myobundles support cellular adaptation in microgravity

We next evaluated endpoint analysis (day 21) for myobundles cultured in microgravity versus ground conditions. To aid in visualizing overall effects across treatment groups, we provide a hierarchical cluster heatmap that depicts DEG-related findings across samples and experimental conditions (Figure S4). Global gene changes are presented as volcano plots, with pathway enrichment terms categorized into BPs, CCs, and MFs. In YA-derived myobundles, E-Stim induced nearly twice as many DEGs compared to No E-Stim samples (1,820 in E-Stim vs. 646 in No E-Stim) (Figures 3A and 3B). These changes were associated with enriched pathways related to RNA binding, translation, mitochondrial function, and muscle contraction (Figures 3C and 3D). Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis identified the focal adhesion pathway as the pathway with the most decreased activity, while GO terms highlighted reductions in cell adhesion, extracellular matrix, and protein binding across both E-Stim and No E-Stim groups. These findings suggest that E-Stim partially mitigates microgravity-induced gene expression changes, although pathways related to extracellular matrix and adhesion remain susceptible to decreased expression under microgravity conditions.

OS-derived myobundles displayed a contrasting pattern compared to YA-derived myobundles. E-Stim under microgravity reduced the total number of DEGs compared to No E-Stim (Figures 4A and 4B). Increased DEGs in both E-Stim and No E-Stim OS groups were associated with cytosol and ATP binding. However, cell adhesion was exclusively enriched in the E-Stim group (Figures 4C and 4D). KEGG analysis identified the hypertrophic cardiomyopathy pathway as enriched in the E-Stim group. Decreased DEGs in OS-derived myobundles were linked to extracellular regions and extracellular matrix structural constituents. Muscle contraction and muscle myosin complex pathways were uniquely de-

creased in No E-Stim samples but remained unaffected in the E-Stim group. KEGG pathway analysis also revealed downregulation of the insulin signaling in No E-Stim samples and upregulation of hypertrophy in E-Stim samples.

These results suggest that E-Stim supports adaptation to microgravity in OS-derived myobundles by mitigating transcriptional response associated with DNA damage and supporting hypertrophy pathways. We identified a limited number of senescence-related genes modulated under microgravity, and most genes were downregulated across the treatment groups suggesting that senescence was not triggered under microgravity conditions (Figure S5). Furthermore, E-Stim appears to promote transcriptional programs linked to cell adhesion and plasma membrane dynamics that support muscle function, particularly in the OS cohort. These findings underscore the differential impact of E-Stim on gene expression in YA- versus OS-derived myobundles under microgravity, with E-Stim showing a potential protective effect against microgravity-induced cellular dysfunction.

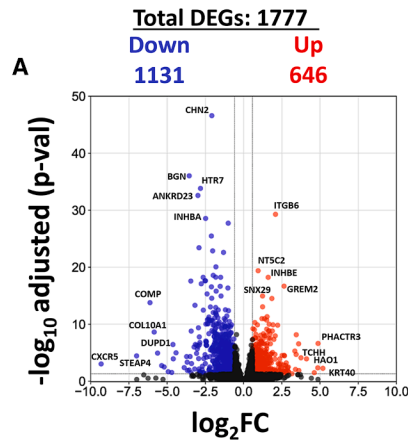
Mitochondrial pathways are altered in E-Stim YA and OS myobundles in microgravity

The impacts of microgravity on mitochondrial function have been well documented in prior studies (Da Silveira et al., 2020; Feger et al., 2016; Nguyen et al., 2021). Our RNA-seq analysis revealed significant changes in gene expression related to mitochondrial energetics under microgravity, with distinct responses observed between YA and OS cohorts. In the YA cohort subjected to E-Stim, 66 DEGs related to mitochondrial function were increased. These included genes involved in oxidative phosphorylation, proton-transporting ATP synthase, mitochondrial electron transport, cytochrome *c* oxidase, NADH dehydrogenase/ubiquinone, and outer mitochondrial membrane organization. Conversely, decreased DEGs were primarily linked to the phosphatidylinositol 3-kinase signaling pathway, which stimulates cytoskeletal arrangement and cell survival (Figure 5A).

In contrast, the OS cohort exhibited upregulation of 20 DEGs regardless of E-Stim treatment. These genes encode for proteins involved in mitochondrial fusion and dynamics related to voltage-gated calcium channels, nucleotide binding and ion transport, and reactive oxygen species regulation (Figure 5A). However, decreased DEGs were associated with antioxidant and mitochondrial enzyme activities including glutathione transfer, superoxide dismutase (*SOD1*), NADH dehydrogenase, and cytochrome oxidase. Notably, activating transcription factor 4 (*ATF4*), a gene associated with muscle atrophy (Adams et al., 2017), was uniquely decreased in E-Stim OS-derived myobundles. Since other muscle atrophy-related genes were not modulated, a decrease in *ATF4* expression is more likely due to



DEGs: Microgravity vs Gravity in YA, No E-Stim



DEGs: Microgravity vs Gravity in YA, E-Stim

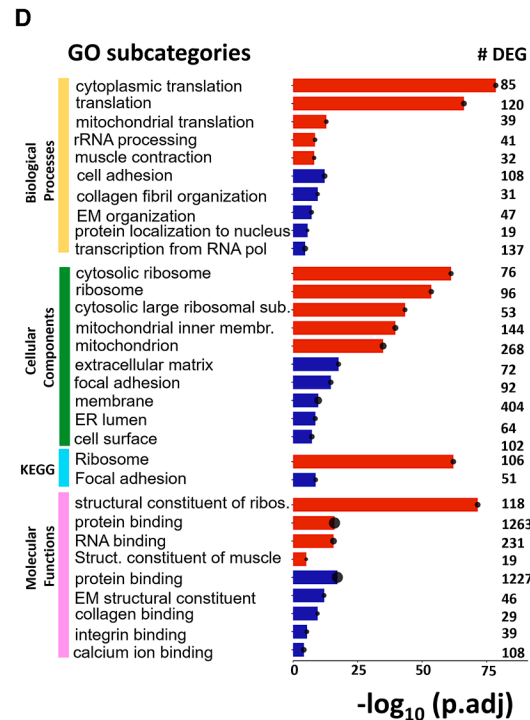
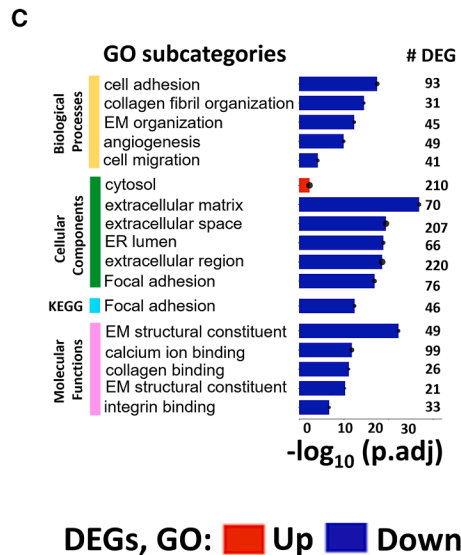
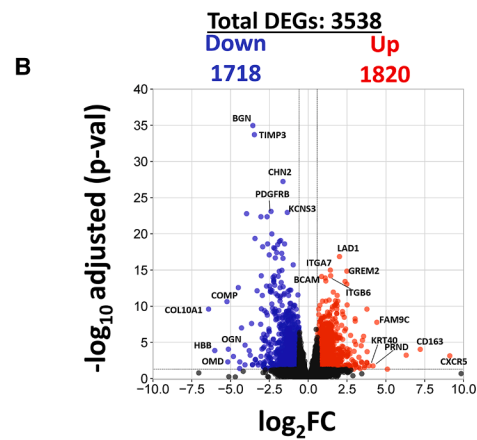


Figure 3. Transcriptome analysis of YA-derived myobundles exposed to microgravity vs. ground

(A and B) Volcano plots of RNA-seq comparing YA-derived myobundles in microgravity vs. ground controls for (A) No E-Stim and (B) E-Stim tissue chips at $\log_2FC$ versus $-\log_{10}$ FDR. Red and blue circles indicated up- and down-regulation of DEGs, respectively, with at least $FDR \leq 0.05$ and $\log_2FC \geq \pm 0.5$ cutoff; black circles denoted non-DEGs.

(C and D) Enriched GO and KEGG analysis between microgravity and gravity control groups for (C) No E-Stim and (D) E-Stim tissue chips. Pathway enrichment analysis by biological processes (yellow), cellular component (green), molecular function (pink), and KEGG (light blue) was performed using DAVID gene functional classification tool.

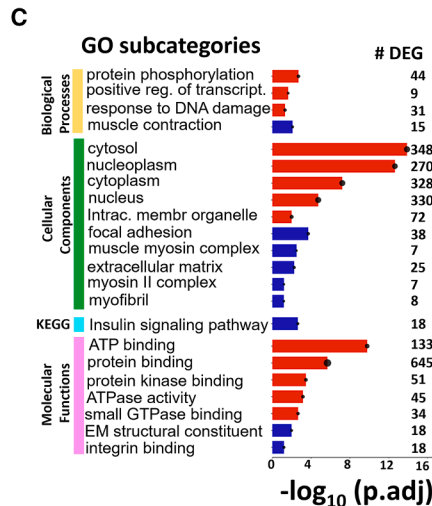
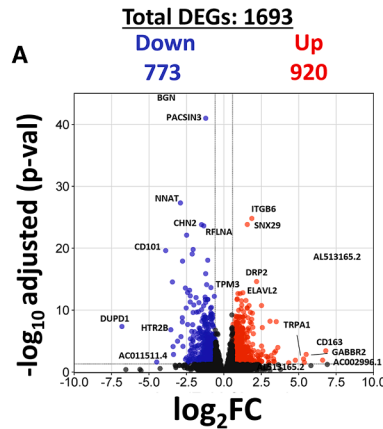
protection against stress response and muscle degradation in microgravity.

Our analysis highlighted a strong signature of mitochondrial energetics in the E-Stim YA-derived myobundles

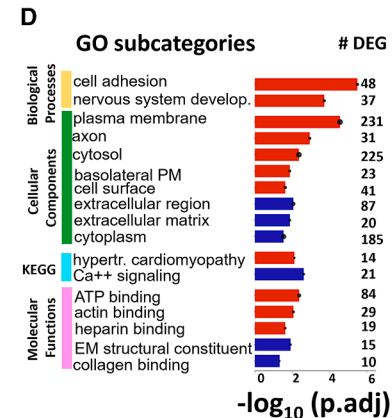
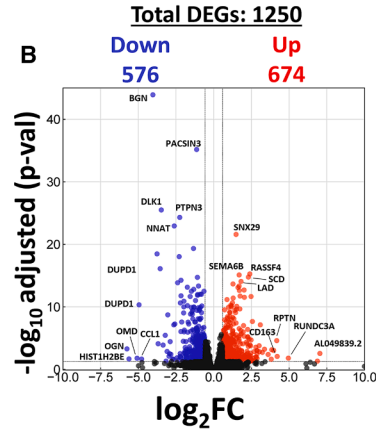
under microgravity compared to ground controls. As depicted in Figure 5B, the expression of genes encoding components of the respiratory chain complexes involved in oxidative phosphorylation was increased. These DEGs



DEGs: Microgravity vs Gravity in OS, No E-Stim



DEGs: Microgravity vs Gravity in OS, E-Stim



DEGs, GO: ■ Up ■ Down

Figure 4. Transcriptome analysis of OS-derived myobundles exposed to microgravity vs. ground

(A and B) Volcano plots of RNA-seq results comparing OS-derived myobundles in microgravity vs. ground controls for (A) No E-Stim and (B) E-Stim tissue chips at $\log_2FC$ versus $-\log_{10}$ FDR. Red and blue circles indicated up- and down-regulation of DEGs, respectively, with at least $FDR \leq 0.05$ and $\log_2FC \geq 0.5$ cutoff; black circles denoted non-DEGs.

(C and D) Enriched GO and KEGG analysis between microgravity and gravity control groups for (C) No E-Stim and (D) E-Stim tissue chips. Pathway enrichment analysis by biological processes (yellow), cellular component (green), molecular function (pink), and KEGG (light blue) was performed using DAVID gene functional classification tool.

included subunits of complex I (mitochondrial-encoded NADH dehydrogenase), complex III, complex IV (cytochrome *c* oxidase subunits), and complex V (ATP synthase). These results underscore the critical role of mitochondrial respiratory chain complexes in adapting to microgravity conditions in E-Stim-treated YA-derived myobundles.

Functional decline of young and older myobundles following microgravity exposure

We evaluated the contractile response of our skeletal muscle tissue chips by recording videos before and during

delivery of exercise-like electrical stimulation. Representative images of YA- and OS-derived myobundles in microgravity and on Earth are shown in Figure 6A, captured inside the CubeLab. Contractile magnitude was determined by analyzing 40-s video sequences captured during E-Stim inside the CubeLab relative to a reference image taken prior to stimulation. Representative videos are included as supplementary material as follows: Video S1 (YA, gravity), Video S2 (OS, gravity), Video S3 (YA, microgravity), and Video S4 (OS, microgravity). Functional assessments were based on the average contractile response of multiple

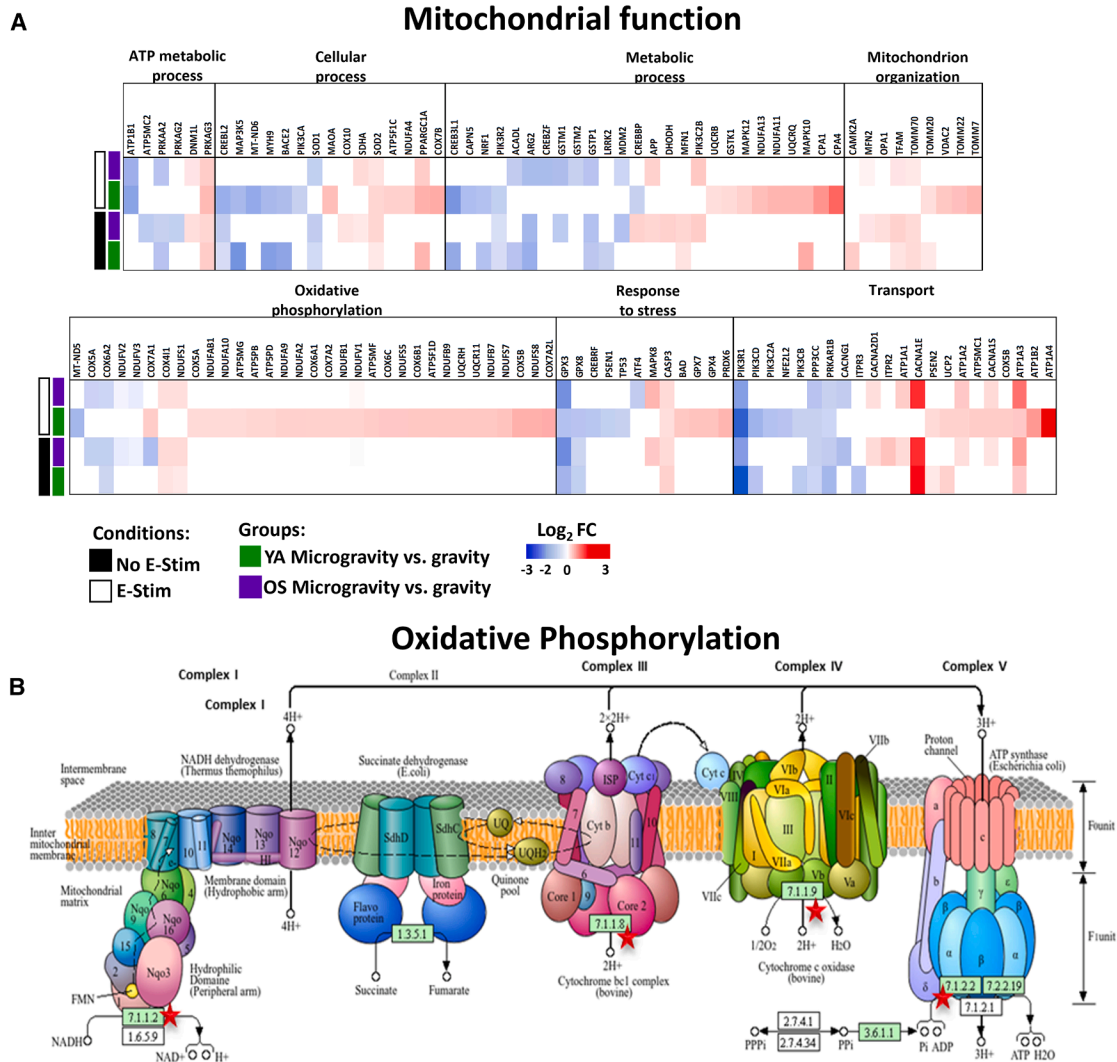


Figure 5. YA E-Stim group shows increased oxidative phosphorylation genes in microgravity

(A) Heatmap of DEGs related to mitochondrial function and their expression intensity across the treatment groups under microgravity vs. gravity.

(B) Respiratory chain complexes involved in oxidative phosphorylation (OXPHOS) identified by KEGG pathway analysis. The red star indicates modulated OXPHOS genes grouped by mitochondrial complexes in response to microgravity.

E-Stim events per tissue chip over 7 days. Displacement measurements plotted over time displayed two peaks per second for electrically synchronized muscle contractions stimulated at 2 Hz frequency (Figure 6B). On Earth, YA-derived myobundles exhibited higher contractile magnitude ($10.4 \pm 3.2 \mu\text{m}$) compared to OS-derived myobundles ($5.6 \pm 1.9 \mu\text{m}$) (Figure 6C). CubeLab data aligned closely with results from ground-based incubator experiments (Figure S1), with YA-derived myobundles demonstrating greater variability in contractile magnitude. In microgravity, contractile function declined, with average contractile magnitudes of $7.7 \pm 4.6 \mu\text{m}$ for YA-derived myobundles and $4.7 \pm 1.0 \mu\text{m}$ for OS-derived myobundles (Figure 6C).

The difference in contractile magnitude values between YA and OS on ground was statistically different; however, the values were not significantly different in microgravity indicating a decline in function in microgravity. Our results align with published reports of human stem cell-derived cardiomyocytes showing a reduction in contractile amplitude after exposure to microgravity (Mair et al., 2024; An et al., 2011; Song et al., 2022).

E-Stim enhances fiber type adaptations and mitochondrial biogenesis in microgravity

MYHs are the major structural protein, integral to the function of striated muscles. Heatmap analysis of muscle fiber

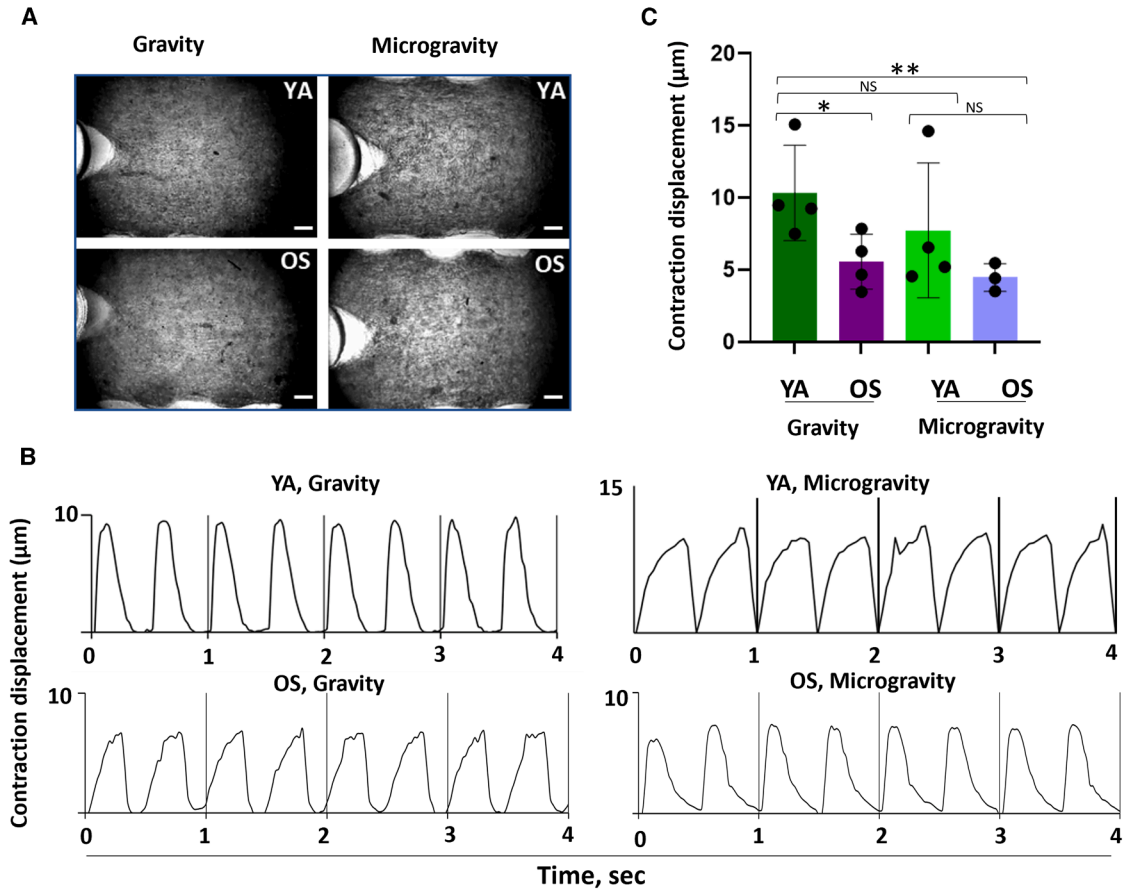


Figure 6. Decreased trend in contraction magnitude in donor-derived myobundles in microgravity

(A) 10× representative photographs of donor-derived myobundles in gravity and microgravity conditions. (B) Contraction displacement traces over time. (C) Contraction displacement magnitude (μm) between young and old myobundles cultured for 12 days. E-Stim regime was applied for 30 min with 12 h rest for 7 days. Video was recorded for 40 s at 40 fps. Myobundles cultured without E-Stim showed contraction displacement values close to zero. Each data point is an average of two recordings per tissue chip.

Statistical significance was calculated using the two-tailed, unpaired Student's t test: YA gravity vs. OS gravity, $*p \leq 0.046$; YA gravity vs. OS microgravity, $**p \leq 0.032$; YA gravity vs. YA microgravity and YA microgravity vs. OS microgravity, NS (not significant), $p > 0.05$.

type-associated gene expression revealed increased expression of genes encoding myosin light and heavy chains, as well as troponin isoforms in E-Stim YA-derived myobundles under microgravity compared to ground controls (Figure 7A). This expression pattern is consistent with type 1, slow-twitch muscle phenotype. E-Stim of both YA and OS cohorts in microgravity vs. gravity upregulated the expression of *MYH7*, a marker of slow-twitch muscle. We also observed increased expression of *PGC-1α*, a transcriptional co-activator that promotes mitochondrial biogenesis and oxidative metabolism, in YA-derived myobundles, under microgravity conditions. The finding aligns with promotion of slow-twitch muscle phenotype, as *PGC-1α* plays a central role in type 1 fiber specification (Lin et al., 2002). *SOX6*, a transcriptional repressor of slow-twitch genes, showed reduced activity across treat-

ment groups, further supporting this phenotype shift (An et al., 2011; Song et al., 2022). Interestingly, the fast-twitch fiber marker *ACTN3* was also increased in the YA E-Stim group, suggesting a mixed fiber-type composition in this cohort.

To confirm the presence of the type MYH7 protein, we performed western blot analysis on the fourth chip from each cohort (protein quantities listed in Table S1) and detected MYH7 across all treatment groups. However, a downward trend in MYH7 protein expression was observed in both YA and OS cohorts under microgravity (Figures 7B and 7C). Statistical analysis comparing averaged MYH7 levels in microgravity versus gravity ($n = 4$) confirmed a significant decrease in MYH7 protein in microgravity-exposed myobundles (Figure 7D). These results suggest that despite transcriptional upregulation, MYH7 protein levels are

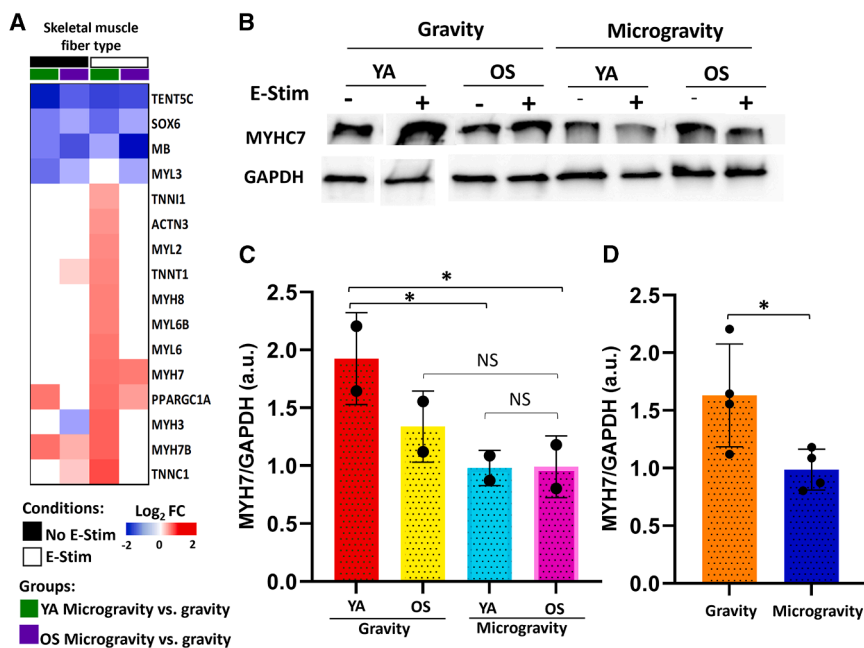


Figure 7. Microgravity alters fiber type I composition in myobundles

(A) Heatmap of expression levels for muscle fiber type I genes between microgravity and gravity in the absence or presence of E-Stim. The expression variance for each gene is indicated by color key ranging from low (blue) to high (red). White denotes no changes in gene expression.

(B) Relative expression levels of MYH7 and GAPDH determined using western blot quantification across cohorts and conditions. The relative expression levels of MYH7 were expressed as the ratio with GAPDH in the same lane.

(C and D) Statistical significance of changes in MYH7 for (C) YA and OS and (D) microgravity vs. gravity using the one-tailed, equal variance Student's t test: * $p \leq 0.05$.

reduced under microgravity, possibly due to increased protein degradation or impaired translational efficiency.

DISCUSSION

Understanding the impact of microgravity on muscle cell biology is critical to developing therapeutic countermeasures to prevent deconditioning and accelerated aging of skeletal muscle during spaceflight. We utilized the CubeLab for controlled and autonomous experiments and evaluated the effects of microgravity on myobundle contractile function and cellular mechanisms through global transcriptome profiling. Our muscle lab-on-chip approach enabled direct comparison between young- and older-derived myobundles in microgravity compared to gravity, highlighting a decline in contractile displacement magnitude, an enrichment of gene pathways associated with mitochondrial function, and a decrease in MYH7 protein levels. As most studies investigating the impact of microgravity on muscle atrophy have been conducted in tissues isolated from humans and rodents in space, our 3D muscle microphysiological system offers the unique opportunity to interrogate the muscle cell's autonomous response to microgravity without the influence of tissue innervation, circulation, and immune cell interactions.

Our spaceflight experiment advances technology readiness of electrically stimulated contraction of human skeletal muscle 3D cells aboard the ISS from our previous mission. We observed reduced contractile magnitude in microgravity compared to ground-based controls. Similar

findings have been reported in a heart-on-chip system, where exposure to microgravity resulted in increased inflammatory signaling and activation of oxidative stress pathways and diminished twitch force and sarcomere organization (Mair et al., 2024). This suggests that attenuated contractile output in microgravity-exposed myobundles may be linked to downregulation of cytoskeletal remodeling pathways, contributing to impaired sarcomere integrity and reduced expression of mitochondrial respiratory chain components. However, in our study, electrically stimulated myobundles, particularly the YA E-Stim samples, show significant modulation of mitochondrial transcription, including increased expression of oxidative phosphorylation genes. On Earth, exercise-like electrical stimulation is known to promote mitochondrial biogenesis and enhance oxidative capacity. To determine whether electrical stimulation can effectively preserve mitochondrial gene expression in microgravity requires future missions incorporating electrical stimulation of muscle cells and live sample return to assess mitochondrial respiration.

Performing experiments in an autonomous CubeLab has the advantage of minimizing crew time and advancing miniaturization and turnkey technologies for evaluating cell responses to different stimuli in normal and extreme environments and over long durations. However, sample size is limited by volume capacity in the CubeLab. We recognize the importance of studying muscle cells from both males and females in the trajectories of sarcopenia (De Jong et al., 2023; Kerr et al., 2024); however, it was not possible to include sufficient samples to robustly compare sex differences. For this ISS study, the optimal approach was to



study one sex. Future work will include a more comprehensive assessment of sex-dependent responses of muscle cells to microgravity and E-Stim. Another limitation of our model is that electrical stimulation alone is not sufficient proxy for mechanical loading experienced by muscle tissue *in vivo*. Performing the experiment in an automated payload paves the way to studying more complex skeletal muscle force assessment in space.

As a final assessment, we directly compared the YA versus the OS groups under microgravity and ground conditions to assess age-related transcriptional differences. Using a database of muscle-aging genes derived from a meta-analysis of 2,852 public gene expression arrays in skeletal muscle (Su et al., 2015), we identified genes affected by microgravity by cross-referencing over 900 genes modulated in human aged skeletal muscle with our dataset. Differential expression analysis between the YA and OS groups is presented as a hierarchical clustering map (Figure S6).

The E-Stim YA vs. E-Stim OS comparison in microgravity exhibited the greatest change in expression. 45 aging genes were significantly increased while a further 41 aging genes were significantly decreased. We observed an increase in aging genes related to inflammation and immune response (e.g., *ALOX5AP*, *CCR10*, and *CFD*), cell signaling and growth (e.g., *WISP2*, *IRS2*, and *SIRT1*), RNA processing (e.g., *CLK4*, *IFRD1*, and *DUSP1*), cellular structure and cytoskeleton regulation (e.g., *ABRA*, *AHNAK*, and *MYL6B*), and metabolism and stress response (e.g., *ASS1*, *SAT1*, *AKRIC3*, and *TRMT112*). *ALOX5AP* expression, which encodes the 5-lipoxygenase-activating protein crucial for activating ALOX5, increased by 2.5-fold in E-Stim YA vs. E-Stim OS under microgravity. This protein has been shown to be increased in aged human skeletal muscle and following glucocorticoid treatment (Kim et al., 2022). However, increased expression of *WISP2*, *IRS2*, and *SIRT1* suggests enhanced muscle regeneration, improved insulin sensitivity and resistance to oxidative stress, and adaptive response in the electrically stimulated YA cohort under microgravity (Powers and Schrager, 2022; Yuan et al., 2016).

Our data revealed a decrease in aging gene expression related to inflammation and immune response (e.g., *TLR4*, *SERPING1*, and *LAMP1*), autophagy (e.g., *SIDT2* and *TECPR2*), cellular stress and senescence (e.g., *TRIM8*, *WARS*, and *AKAP12*), and extracellular matrix remodeling (e.g., *ADAMTSS* and *SPOCK1*). We observed that a decrease in *FOXO1*, a key transcription factor in the insulin/IGF-1 signaling pathway, may indicate the suppression of muscle protein breakdown during E-Stim. Furthermore, we observed a 2-fold decrease in Toll-like receptor 4 (*TLR4*) expression. Previous studies have shown that stimulated myotube contractions reduce membrane-bound TLR4 while increasing soluble TLR4, thereby inhibiting TLR4-mediated

signaling and preventing myotube atrophy (Ducharme et al., 2023).

In conclusion, our study employed morphological, physiological, and transcriptomic approaches to investigate the adaptation of donor-derived skeletal muscle myobundles to microgravity. Our findings highlight significant transcriptional differences between young and old muscle phenotypes under microgravity, providing insights into how microgravity influences or accelerates age-related changes in skeletal muscle, and highlight the potential role of electrical stimulation in partially mitigating tissue degradation.

METHODS

Donor and muscle biopsy data analysis

Cells from the same lot of muscle biopsy cells used for our SpaceX CRS-21 mission (Parafati et al., 2023) were subsequently purified and seeded for the SpaceX CRS-25 mission and represented independent experiments. In brief, percutaneous biopsies were obtained from the middle region of the vastus lateralis using a Bergstrom needle (as previously described, Sparks et al., 2011), from five young, active male volunteers (mean 35 ± 4) and five old, sedentary male volunteers (mean 70 ± 4). Participants were considered active if they engaged in endurance exercise at least 3 days a week without extensive lay off over the previous 6 months. Participants were considered sedentary if they completed one or fewer structured exercise sessions a week. The study protocol was reviewed and approved by the institutional review board at AdventHealth, Orlando (IRBNet #554559).

Myoblast purification

Harvested cells were purified with an immuno-magnetic sorting system (Miltenyi Biotec, Paris, France) using an anti-CD56/NCAM antibody as described (Giza et al., 2022). Purified myoblasts were plated in collagen-coated Petri dishes and cultured in skeletal muscle cell growth medium (PromoCell Cat# C-23060) supplemented with 0.1 mg/mL primosin to protect cultures from microbial contamination, 5% fetal bovine serum, and gentamicin at 37°C in a humidified atmosphere with 5% CO₂. At cell isolation, all human myoblasts were at passage 4 and expanded to passage 6. Passage numbers were matched for young- and old-derived donor cells. Myoblasts from different donors were pooled in a 1:1 ratio, and aliquots were cryopreserved. Enrichment was confirmed by comparing CD56⁺ and CD56⁺ cell populations by fluorescence-activated cell sorting staining as described (Giza et al., 2022).

Myobundle culture and maintenance in CubeLab

Donor-derived 3D myobundles were generated by seeding myoblasts in tissue-engineered constructs and launched to



the ISS aboard the SpaceX CRS-25 commercial resupply mission. Cell seeding was performed at NASA's Kennedy Space Station Processing Facility (SSPF) for flight and asynchronously at the University of Florida laboratory for the ground test. CD56⁺-enriched myoblasts were mixed in a collagen-Matrigel hydrogel and injected into the PDMS-based chips at a final cell density of 15 and 20 million cells/mL, for YA and OS-derived cell stocks, respectively, and allowed to polymerize at 37°C for 60 min as previously described (Giza et al., 2022; Parafati et al., 2023).

Differentiation followed a two-step process as previously described (Giza et al., 2022; Parafati et al., 2023, 2025). Sixteen tissue chips were integrated onto the CubeLab manifold on day 8 of growth, and media for flight was stored at 6°C–8°C in a gas-permeable bag. On day 14, 30 min of E-Stim of 3 V, 2 Hz, 2-ms pulse was applied sequentially to four YA and four OS tissue chips; the other eight chips were No E-Stim. Donor-derived 3D myobundles were maintained in the Space Tango CubeLab at 37°C and 5%–6% CO₂ for both the flight and ground payloads. The myobundles were cultured in maintenance medium that was changed every 6 h to maintain healthy muscle microtissues.

Measurement of myotube contractile activity

Myotube contractility was quantified by DIC. The DIC analysis was done by tracking the regional variation of grayscale features between two succession images recorded before and after E-Stim by an optics system installed in the CubeLab. Images are acquired with a resolution of 3,088 × 2,076 pixels (2.6 × 1.7 mm). A pixel corresponds to 0.84 μm. Image processing was similarly described (Giza et al., 2022). In brief, 40-s videos captured at 40 fps were extracted for a sequence of 1,601 tiff images using MATLAB software. DIC was performed using GOM correlate professional software (Zeiss, Germany). The reference image (the first frame prior to application of E-Stim) is compared to each test image (i.e., with E-Stim applied) in the sequence. The DIC values are saved as an Excel file that stores the numerical displacement info for each grid at individual frames.

RNA isolation and preparation

We performed transcript profiling of YA- and OS-derived myoblasts cultured for 2 and 21 days (including 12 days of spaceflight) and compared them with myoblasts grown on ground for the full 21 days by RNA-seq. Cultures in tissue chip devices were preserved with RNAlater (Thermo Fisher Scientific) and frozen at –20°C to –30°C. Cells were lysed in the chip, and RNA isolation was performed using RNeasy Plus Mini Kit (QIAGEN) as previously described (Parafati et al., 2023). RNA concentration and quality were determined with Agilent TapeStation 2200 (Agilent Technologies, Santa Clara, CA, USA). 500 ng of total

RNA was used for RNA sample preparation and processed in triplicate for a total of 30 samples. The sequencing generated over 50 million unique mapped reads per library from six RNA-seq libraries constructed from YA- and OS-derived myoblasts prior to myogenic differentiation and 24 RNA-seq libraries constructed from E-Stim and No E-Stim YA- and OS-derived myobundles on day 21.

RNA-sequencing library preparation and RNA-seq data processing

Strand-specific RNA-seq libraries were generated using the NEBNext Ultra Directional RNA Library Prep Kit for Illumina (NEB, USA) as previously described (Parafati et al., 2023). Differential expression analysis was performed using the DESeq2 package, with a false discovery rate (FDR)-corrected *p* value threshold of 0.05. Sequencing and bioinformatics were performed at the University of Florida Interdisciplinary Center for Biotechnology Research core facility.

Functional enrichment analysis of DEGs

Functional enrichment of gene datasets was performed using GO analysis (Huang et al., 2009). GO enrichment was categorized into BP, MF, and CC classes. To further interpret genomic information, KEGG pathway enrichment analysis was performed (Kanehisa and Goto, 2000). Both GO annotation and KEGG pathway enrichment analysis were carried out using bioinformatics tools from the Database for Annotation, Visualization, and Integrated Discovery (DAVID) Gene Functional Classification Tool, <https://david.ncifcrf.gov/>. GO and KEGG analysis were performed using genes that were differentially expressed with FDR ≤ 0.05 and log₂ fold change (FC) cutoff of ≥ ±0.5 or ≥ ±1.5. For GO analysis, enriched terms were considered significant with a corrected *p* value ≤ 1E–6 and an uncorrected *p* value ≤ 0.01. For KEGG analysis, a corrected enrichment *p* value ≤ 0.05 was applied. Only pathways with gene counts > 5 were considered significantly enriched. The annotation dataset was set to PANTHER Protein Class, and analysis was performed using Fisher's exact test with FDR correction to account for multiple comparisons.

Sarcomere immunostaining and confocal microscopy

The confocal microscopy was performed according to the method previously reported (Giza et al., 2022). Myobundles images were acquired using a Biotek Cytation C10 Imager (Agilent, Biotek) with 20× objectives.

MYH western blot analysis

Myobundle proteins were isolated in RIPA lysis buffer (Cell Signaling Technology) with protease inhibitors (Thermo Fisher Scientific). Protein concentration was determined using bicinchoninic acid colorimetric assay kits (Thermo Fisher Scientific). Western blot was performed using



4%–12% SDS-PAGE gel Bio-Rad Mini-PROTEAN gels (Bio-Rad, Hercules, CA) and transferred onto a polyvinylidene difluoride membrane (Bio-Rad). The primary myosin heavy chain MYH7 (1:1,000) antibody and HRP-conjugated anti-mouse and anti-rabbit antibodies (1:5,000) were purchased from Cell Signaling Technology. GAPDH (1:2,000) was used as an internal control. Chemiluminescence was performed using Pierce ECL Western Blotting Substrate (Thermo Fisher Scientific). Images were acquired using a Chemidoc (Bio-Rad) and analyzed using NIH ImageJ (<https://imagej.net/ij/>).

Statistical analysis

DEGs were detected by setting the adjusted p value (p_{adj}) to ≤ 0.05 . All quantitative data were analyzed and plotted using GraphPad Prism 8.4.3 (GraphPad Software, Inc.). Data are presented as the mean $\pm$ standard deviation. Fisher's exact test was applied to assess gene enrichment in annotation terms and identify overrepresented pathways with significant overlaps within predefined gene sets. p values were computed by summing probabilities over defined sets of tables ($\text{Prob} = \sum A_p$) to evaluate functional enrichment against a background of all *Homo sapiens* genes. Significance of each GO term was determined using the default *Homo sapiens* GO annotation as background and considered statistically significant if the FDR-corrected p value was $\leq 1E-6$ and uncorrected p value was ≤ 0.01 . DEGs were significant according to $\log_2\text{FC}$ with threshold set at ± 1.5 and $*p$ value ≤ 0.05 (Bonferroni-corrected p value). Data are representative of three independent RNA-seq determinations. For volcano plots, bar dots were ordered according to the significance and are ranked by $\log_{10}$ FDR-adjusted p values. The horizontal axis represents the $\log_{10}$ adjusted p values of enriched DEGs to background genes, while the vertical axis represents the name of enriched sub-categories. Numbers in parentheses indicate enriched DEGs reported in each sub-ontology category.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to the lead contact, Siobhan Malany (smalany@ufl.edu).

Materials availability

This study did not generate new unique reagents.

Unless otherwise noted, data are available in the main text or the supplementary materials.

Data and code availability

The accession number for the RNA-sequencing data reported here is Gene Expression Omnibus: GSE298393. Donor-derived cells are obtained through a materials transfer agreement and based on availability.

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AUTHOR CONTRIBUTIONS

M.P. and S.M. conceived and designed the project. M.P. and S.M. performed and analyzed most experiments with assistance from Z.T. and P.M.C. M.P. and S.M. performed experiments at Kennedy Space Center SSPF labs with assistance from L.K.M. P.M.C. led the clinical study from which donor-derived myoblasts were derived. M.P. conducted bioinformatics, viability, staining, and western blotting analysis. M.P. performed DIC analysis with assistance from Z.T. S.M. supervised the project and acquired funding. S.M. and M.P. wrote the manuscript with input from all authors. S.M., M.P., and P.M.C. edited the manuscript.

DECLARATION OF INTERESTS

S.M. is the founder of Micro-gRx and a member of the scientific board.

L.K.M. is the author of a patent application for the microfluidic device.

SUPPLEMENTAL INFORMATION

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REFERENCES

Adams, C.M., Ebert, S.M., and Dyle, M.C. (2017). Role of ATF4 in skeletal muscle atrophy. *Curr. Opin. Clin. Nutr. Metab. Care* 20, 164–168. <https://doi.org/10.1097/MCO.0000000000000362>.



- An, C.-I., Dong, Y., and Hagiwara, N. (2011). Genome-wide mapping of Sox6 binding sites in skeletal muscle reveals both direct and indirect regulation of muscle terminal differentiation by Sox6. *BMC Dev. Biol.* 11, 59. <https://doi.org/10.1186/1471-213X-11-59>.
- Cannavo, A., Carandina, A., Corbi, G., Tobaldini, E., Montano, N., and Arosio, B. (2022). Are Skeletal Muscle Changes during Prolonged Space Flights Similar to Those Experienced by Frail and Sarcopenic Older Adults? *Life* 12, 2139. <https://doi.org/10.3390/12122139>.
- Comfort, P., McMahon, J.J., Jones, P.A., Cuthbert, M., Kendall, K., Lake, J.P., and Haff, G.G. (2021). Effects of Spaceflight on Musculoskeletal Health: A Systematic Review and Meta-analysis, Considerations for Interplanetary Travel. *Sports Med.* 51, 2097–2114. <https://doi.org/10.1007/s40279-021-01496-9>.
- Da Silveira, W.A., Fazelinia, H., Rosenthal, S.B., Laiakis, E.C., Kim, M.S., Meydan, C., Kidane, Y., Rathi, K.S., Smith, S.M., Stear, B., et al. (2020). Comprehensive Multi-omics Analysis Reveals Mitochondrial Stress as a Central Biological Hub for Spaceflight Impact. *Cell* 183, 1185–1201.e20. <https://doi.org/10.1016/j.cell.2020.11.002>.
- De Jong, J.C.B.C., Attema, B.J., Van Der Hoek, M.D., Verschuren, L., Caspers, M.P.M., Kleemann, R., Van Der Leij, F.R., Van Den Hoek, A.M., Nieuwenhuizen, A.G., and Keijer, J. (2023). Sex differences in skeletal muscle-aging trajectory: same processes, but with a different ranking. *GeroScience* 45, 2367–2386. <https://doi.org/10.1007/s11357-023-00750-4>.
- Ducharme, J.B., Fennel, Z.J., McKenna, Z.J., Nava, R.C., and Deyhle, M.R. (2023). Stimulated myotube contractions regulate membrane-bound and soluble TLR4 to prevent LPS-induced signaling and myotube atrophy in skeletal muscle cells. *Am. J. Physiol. Cell Physiol.* 325, C300–C313. <https://doi.org/10.1152/ajpcell.00007.2023>.
- Feger, B.J., Thompson, J.W., Dubois, L.G., Kommaddi, R.P., Foster, M.W., Mishra, R., Shenoy, S.K., Shibata, Y., Kidane, Y.H., Moseley, M.A., et al. (2016). Microgravity induces proteomics changes involved in endoplasmic reticulum stress and mitochondrial protection. *Sci. Rep.* 6, 34091. <https://doi.org/10.1038/srep34091>.
- Francetic, T., and Li, Q. (2011). Skeletal myogenesis and Myf5 activation. *Transcription* 2, 109–114. <https://doi.org/10.4161/trns.2.3.15829>.
- Giza, S., Mojica-Santiago, J.A., Parafati, M., Malany, L.K., Platt, D., Schmidt, C.E., Coen, P.M., and Malany, S. (2022). Microphysiological system for studying contractile differences in young, active, and old, sedentary adult derived skeletal muscle cells. *Aging Cell* 21, e13650. <https://doi.org/10.1111/accel.13650>.
- Huang, D.W., Sherman, B.T., and Lempicki, R.A. (2009). Systematic and integrative analysis of large gene lists using DAVID bioinformatics resources. *Nat. Protoc.* 4, 44–57. <https://doi.org/10.1038/nprot.2008.211>.
- Juhl, O.J., Buettmann, E.G., Friedman, M.A., DeNapoli, R.C., Hoppeck, G.A., and Donahue, H.J. (2021). Update on the effects of microgravity on the musculoskeletal system. *NPJ Microgravity* 7, 28. <https://doi.org/10.1038/s41526-021-00158-4>.
- Kanehisa, M., and Goto, S. (2000). KEGG: kyoto encyclopedia of genes and genomes. *Nucleic Acids Res.* 28, 27–30. <https://doi.org/10.1093/nar/28.1.27>.
- Kerr, H.L., Krumm, K., Anderson, B., Christiani, A., Strait, L., Li, T., Irwin, B., Jiang, S., Rybachok, A., Chen, A., et al. (2024). Mouse sarcopenia model reveals sex- and age-specific differences in phenotypic and molecular characteristics. *J. Clin. Investig.* 134, e172890. <https://doi.org/10.1172/JCI172890>.
- Khodabukus, A., Madden, L., Prabhu, N.K., Koves, T.R., Jackman, C.P., Muoio, D.M., and Bursac, N. (2019). Electrical stimulation increases hypertrophy and metabolic flux in tissue-engineered human skeletal muscle. *Biomaterials* 198, 259–269. <https://doi.org/10.1016/j.biomaterials.2018.08.058>.
- Kim, H.-J., Kim, S.-W., Lee, S.-H., Jung, D.-W., and Williams, D.R. (2022). Inhibiting 5-lipoxygenase prevents skeletal muscle atrophy by targeting organogenesis signalling and insulin-like growth factor-1. *J. Cachexia Sarcopenia Muscle* 13, 3062–3077. <https://doi.org/10.1002/jcsm.13092>.
- Krittanawong, C., Singh, N.K., Scheuring, R.A., Urquieta, E., Bershad, E.M., Macaulay, T.R., Kaplin, S., Dunn, C., Kry, S.F., Russomano, T., et al. (2022). Human Health during Space Travel: State-of-the-Art Review. *Cells* 12, 40. <https://doi.org/10.3390/cells12010040>.
- Krüger, M., and Kötter, S. (2016). Titin, a Central Mediator for Hypertrophic Signaling, Exercise-Induced Mechanosignaling and Skeletal Muscle Remodeling. *Front. Physiol.* 7, 76. <https://doi.org/10.3389/fphys.2016.00076>.
- Larsson, L., Degens, H., Li, M., Salvati, L., Lee, Y.I., Thompson, W., Kirkland, J.L., and Sandri, M. (2019). Sarcopenia: Aging-Related Loss of Muscle Mass and Function. *Physiol. Rev.* 99, 427–511. <https://doi.org/10.1152/physrev.00061.2017>.
- Lee, P.H.U., Chung, M., Ren, Z., Mair, D.B., and Kim, D.-H. (2022). Factors mediating spaceflight-induced skeletal muscle atrophy. *Am. J. Physiol. Cell Physiol.* 322, C567–C580. <https://doi.org/10.1152/ajpcell.00203.2021>.
- Leuchtmann, A.B., Adak, V., Dilbaz, S., and Handschin, C. (2021). The Role of the Skeletal Muscle Secretome in Mediating Endurance and Resistance Training Adaptations. *Front. Physiol.* 12, 709807. <https://doi.org/10.3389/fphys.2021.709807>.
- Lin, J., Wu, H., Tarr, P., et al. (2002). Transcriptional co-activator PGC-1 α drives the formation of slow-twitch muscle fibres. *Nature* 418, 797–801. <https://doi.org/10.1038/nature0090>.
- Lu, S., E, B.D., and Horowitz, R. (2008). Expression and alternative splicing of N-RAP during mouse skeletal muscle development. *Cell Motil. Cytoskeleton* 65, 945–954.
- Mair, D.B., Tsui, J.H., Higashi, T., Koenig, P., Dong, Z., Chen, J.F., Meir, J.U., Smith, A.S.T., Lee, P.H.U., Ahn, E.H., et al. (2024). Spaceflight-induced contractile and mitochondrial dysfunction in an automated heart-on-a-chip platform. *Proc. Natl. Acad. Sci.* 121, e2404644121. <https://doi.org/10.1073/pnas.2404644121>.
- Nguyen, H.P., Tran, P.H., Kim, K.-S., and Yang, S.-G. (2021). The effects of real and simulated microgravity on cellular mitochondrial function. *Npj Microgravity* 7, 44. <https://doi.org/10.1038/s41526-021-00171-7>.



- Nikolić, N., Bakke, S.S., Kase, E.T., Rudberg, I., Flo Halle, I., Rustan, A.C., Thoresen, G.H., and Aas, V. (2012). Electrical Pulse Stimulation of Cultured Human Skeletal Muscle Cells as an In Vitro Model of Exercise. *PLoS One* 7, e33203. <https://doi.org/10.1371/journal.pone.0033203>.
- Parafati, M., Giza, S., Shenoy, T.S., Mojica-Santiago, J.A., Hopf, M., Malany, L.K., Platt, D., Moore, I., Jacobs, Z.A., Kuehl, P., et al. (2023). Human skeletal muscle tissue chip autonomous payload reveals changes in fiber type and metabolic gene expression due to spaceflight. *Npj Microgravity* 9, 77. <https://doi.org/10.1038/s41526-023-00322-y>.
- Parafati, M., Shenoy, T.S., Thwin, Z., Parlavacchio, M., and Malany, S. (2025). Tomatidine Attenuates Inflammatory and Structural Responses to Exercise-Like Stimulation in Donor-derived Skeletal Muscle Myobundles. *Medical Research Archives* 13, 15.
- Powers, S.K., and Schrager, M. (2022). Redox signaling regulates skeletal muscle remodeling in response to exercise and prolonged inactivity. *Redox Biol.* 54, 102374. <https://doi.org/10.1016/j.redox.2022.102374>.
- Song, M.-Y., Han, C.Y., Moon, Y.J., Lee, J.H., Bae, E.J., and Park, B.-H. (2022). Sirt6 reprograms myofibers to oxidative type through CREB-dependent Sox6 suppression. *Nat. Commun.* 13, 1808. <https://doi.org/10.1038/s41467-022-29472-5>.
- Sparks, L.M., Moro, C., Ukropcova, B., Bajpeyi, S., Civitarese, A.E., Hulver, M.W., Thoresen, G.H., Rustan, A.C., and Smith, S.R. (2011). Remodeling lipid metabolism and improving insulin responsiveness in human primary myotubes. *PLoS One* 6, e21068. <https://doi.org/10.1371/journal.pone.0021068>.
- Su, J., Ekman, C., Oskolkov, N., Lahti, L., Ström, K., Brazma, A., Groop, L., Rung, J., and Hansson, O. (2015). A novel atlas of gene expression in human skeletal muscle reveals molecular changes associated with aging. *Skelet. Muscle* 5, 35. <https://doi.org/10.1186/s13395-015-0059-1>.
- Tidball, J.G. (2005). Mechanical signal transduction in skeletal muscle growth and adaptation. *J. Appl. Physiol.* 98, 1900–1908. <https://doi.org/10.1152/japplphysiol.01178.2004>.
- Trappe, S., Costill, D., Gallagher, P., Creer, A., Peters, J.R., Evans, H., Riley, D.A., and Fitts, R.H. (2009). Exercise in space: human skeletal muscle after 6 months aboard the International Space Station. *J. Appl. Physiol.* 106, 1159–1168. <https://doi.org/10.1152/japplphysiol.91578.2008>.
- Tripathi, A.K., Ramani, U.V., Rank, D.N., and Joshi, C.G. (2011). In vitro expression profiling of myostatin, follistatin, decorin and muscle-specific transcription factors in adult caprine contractile myotubes. *J. Muscle Res. Cell Motil.* 32, 23–30. <https://doi.org/10.1007/s10974-011-9245-x>.
- Valdez, M.R., Richardson, J.A., Klein, W.H., and Olson, E.N. (2000). Failure of Myf5 to support myogenic differentiation without myogenin, MyoD, and MRF4. *Dev. Biol.* 219, 287–298. <https://doi.org/10.1006/dbio.2000.9621>.
- Vernikos, J., and Schneider, V.S. (2010). Space, Gravity and the Physiology of Aging: Parallel or Convergent Disciplines? A Mini-Review. *Gerontology* 56, 157–166. <https://doi.org/10.1159/000252852>.
- Williams, D., Kuipers, A., Mukai, C., and Thirsk, R. (2009). Acclimation during space flight: effects on human physiology. *Can. Med. Assoc. J.* 180, 1317–1323. <https://doi.org/10.1503/cmaj.090628>.
- Yuan, Y., Cruzat, V.F., Newsholme, P., Cheng, J., Chen, Y., and Lu, Y. (2016). Regulation of SIRT1 in aging: Roles in mitochondrial function and biogenesis. *Mech. Ageing Dev.* 155, 10–21. <https://doi.org/10.1016/j.mad.2016.02.003>.
- Zhao, S., Mehta, A.S., and Zhao, M. (2020). Biomedical applications of electrical stimulation. *Cell. Mol. Life Sci.* 77, 2681–2699. <https://doi.org/10.1007/s00018-019-03446-1>.